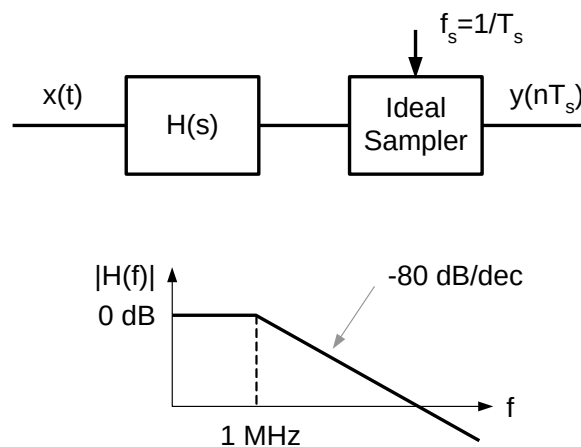


HOMework #1

(Due: Wednesday, February 22, 2023, 12:00 pm PT via Brightspace)

1. **Anti-Alias Filtering.** Consider the system shown below.
 - a) What is the minimum sampling frequency f_s that guarantees > 70 dB attenuation of in-band aliasing components in the discrete time signal $y(nT_s)$? Note that the system is designed for a “user” signal bandwidth of 1 MHz. What is the resulting oversampling ratio (OSR)?
 - b) What is the new required oversampling ratio if the filter roll-off is reduced to -60 dB/decade?

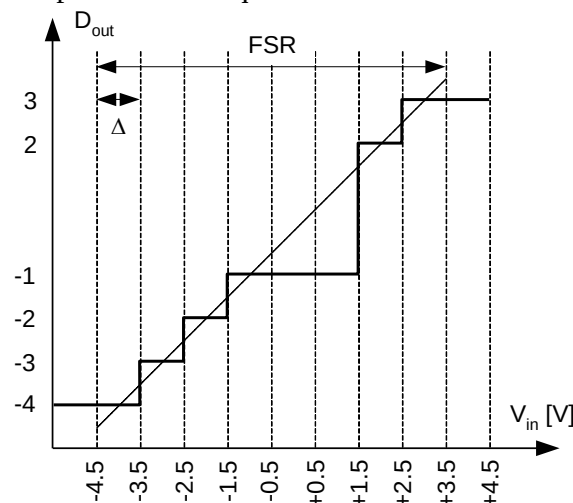


2. **State-of-the-art Data Converters.** In class, we’ve looked at a few examples of recently published high performance A/D converters. You are to find 3 additional journal or conference papers on ADCs published in 2022. For each paper, list the following:
 - a) Complete reference in IEEE format. Example: S.-T. Ryu, B.-S. Song and K. Bacrania, “A 10-bit 50-MS/s Pipelined ADC With Opamp Current Reuse,” *IEEE J. Solid-State Circuits*, vol. 42, no. 3, pp. 475-485, Mar. 2007.
 - b) Stated bit-resolution, sampling rate and power dissipation
 - c) Technology and chip area
 - d) Architecture and application
 - e) Just for fun, calculate how long (in days, hours or minutes) the described chip could be powered from a cell phone battery (even though this may not be the intended application). A typical cell phone battery holds approximately 5 Wh of energy.

Note: You may use e.g. IEEE's publication search engine available at <http://ieeexplore.ieee.org>. Any computer with a TU Delft IP address has full-access to this database.

3. Quantization Noise Power Calculation for a Ramp Input Signal

- Consider an ideal 3-bit mid-tread quantizer with a ramp input that runs from the beginning to the end of the full scale range between $t=0$ and $t=t_1$. Sketch the quantization error as a function of time and calculate its power. The result may or may not surprise you. Why didn't we follow this approach for the sinusoidal example discussed in class?
- We now consider a slightly "broken" quantizer (shown in the diagram below), with the same input. Again, sketch the quantization error over time compute the quantization noise power of this quantizer.



4. Quantization Noise Model Validation. For a uniform quantizer, we know that under certain conditions, the variance (or "power") of the quantization error can be approximated as $\Delta^2/12$. In this problem, we will study this approximation for various resolutions and two different signals.

- Write a Matlab routine that implements a 3-bit mid-rise quantizer with a full-scale range of $-1 \dots +1$. Submit a plot of the quantizer's transfer function and the associated quantization error. There is no need to "encode" the quantizer output in the context of this problem, i.e. the quantizer output will be a real number in the range ± 1 .
- Apply the following signal to the quantizer: $x = \sin(2\pi \cdot 0.107 \cdot n)$, where $n = 1 \dots 10,000$. Compute the mean square value of the quantization error vector and compare to the value of $\Delta^2/12$. What is the percent deviation relative to $\Delta^2/12$?

- c) Repeat part (b) for resolutions of 2...18 bits (e.g., using a for-loop in Matlab). Plot the normalized mean square quantization error across resolution; i.e. x-axis: bit resolution, y-axis: mean square quantization error divided by $\Delta^2/12$.
 - d) Repeat part (c) using 10,000 samples of a random input signal, uniformly distributed in $[-1 +1]$, as the input. In the same diagram, plot the normalized mean square quantization error across resolution, as described in (c).
 - e) Summarize your observations from the results obtained under (c) and (d).
5. **SQNR calculation.** In the system shown below, the quantizer is ideal and has a resolution of 14 bits. The input signal is a periodic full-scale half-wave rectified sinusoid waveform. The samples are taken such that the quantization error is sufficiently randomized (as assumed by the standard statistical quantizer model). Calculate the SQNR (in dB) at the output (D_{out}). Consider only quantization noise.

